

Protect or Perish: The Domain Defense Challenge

Learning Objective

Your goal is to design and implement a cost-effective defensive registration strategy that reduces the likelihood of a company having to engage in future domain disputes against cyber squatters. Using *empirical analysis* (on historical data from several Fortune 500 companies) and/or *qualitative research methods* (e.g., based on interviews or market analysis), you will demonstrate your ability to:

1. Determine which domains related to a company's brand are attractive targets for squatters based on historical data.
2. Recommend which domains a perspective client should defensively register, given a fixed budget for online brand protection.
3. Pitch your quantitative- or qualitatively driven protection strategy to your firm's leadership who intends to sell it to a prominent prospect.

This project is worth 20% of your final course grade.

Important Deadlines

Monday December 1st, 6:00pm	Submit code Notebook / Script. Your code will be run on a new holdout dataset.
Tuesday December 2nd, 12:30pm	In-class presentation, Van Leer C241
Friday December 11th, 6:00pm	Final report

Teaming

You must work in a group of four. Each team member should have a well-defined role (e.g., operations, marketing/sales, visual designer, data analyst, engineer, etc.).

Pitch Perfect Protection: Winning Strategies for Smarter Domain Defense

You are part of the customer acquisition team at an online brand protection (OBP) service provider specializing in safeguarding organizations' digital assets. A major prospect, *Strategic Ventures Inc.*, managing digital operations for ten (10) Fortune 500 companies has issued a Request for Proposals (RFP) to a handful of pre-selected firms, seeking solutions to minimize domain name abuse and future dispute costs under the Uniform Rapid Suspension (URS) policy. The draft contract outlines specific requirements, including a defined budget allocation for domain registrations and a penalty structure that applies at varying rates if disputes arise regarding domains not registered by your firm. A detailed breakdown of these requirements is provided in the following table.

Customer 5-year contract value	\$100,000
5-year registration cost per domain	\$200
Cost per URS dispute	\$1,200
Cost share for failure to defensively register a domain name that ends up being disputed	- \$300 (25%) for <i>low-risk domains</i> - \$600 (50%) for <i>medium-risk domains</i> - \$900 (75%) for <i>high-risk domains</i>

Due to the significant compensation and prestige tied to this contract, the CTO at your firm, *DomainSense*, has accepted the invitation to participate and has tasked several groups with analyzing data from existing customers to develop an evidence-based defensive registration plan that **optimizes the client's budget, prevents future disputes, while ensuring profitability for the firm**. The groups will be evaluated by a panel of expert judges based on two criteria: (1) ***the persuasiveness of their pitch*** and (2) ***the performance of their strategy on a subset of companies for which the dispute data has been withheld***. The winning group will receive a substantial year-end bonus, and their strategy will form the foundation of *DomainSense's* final proposal to the prospect.

Deliverables

1. Submit a well-documented source code that accepts a set of corporate domain names as input and outputs a recommended list of domains for defensive registration within the

client's allocated budget. Your implementation must adhere to the guidelines provided in the starter code shared with you to enable an independent evaluation of your approach on the hold-out dataset.

2. Deliver a 20-minute presentation followed by a Q&A session with a panel of judges. The presentation must justify the chosen strategy and highlight its benefits for the client. You must ensure that your pitch is engaging and contains compelling evidence for the acquisition team to make an informed decision. The acquisition team will report to *DomainSense's* CTO which team's strategy was more convincing and performed well on a hold-out dataset. That team's strategy will be refined to compete against other firms for the ultimate five-year deal.
3. Submit a strategy brochure (4 pages maximum) mentioning the names and roles of each group member. The report should detail your methodology, your insights, the justification for the adopted strategy, and overall recommendation. Use visually appealing graphics to hook your audience. You must also include a concise post-mortem analysis of your performance on the hold-out dataset, explaining why you still believe your approach has value, even in cases where the predictions were inaccurate. After all, the deal isn't set until the contract is signed and delivered :)

The OBP Gauntlet: Strategy Leaderboard

To allow teams to iteratively improve their strategies and assess how well their techniques perform against each other, the CTO set up a leaderboard to view in real-time the performance of each team's strategy on the holdout dataset. The scoring rewards teams whose recommendations lead to fewer domain name disputes while remaining profitable.



Data

All the data needed to complete this project can be found on [Canvas](#) . Please make sure to read the instructions in this document and the **README.md** file in its entirety.

You are provided the following details about each of *DomainSense's* existing clients included in the dataset: the company's name, business sector, market value, financial asset size, popularity ranking of their corporate domain, and open-source intelligence on the frequency of malicious activity affecting their brands. The **companies-training.jsonl** and **companies-validation.jsonl** files provide those details respectively for the companies in the training and validation sets. The validation set refers to the data used to assess the submissions on the public leaderboard.

You are also provided with the entire set of domains disputed by the clients in the training dataset which you can find in the file **disputes-training.json** in the format displayed below. As the screenshot shows, the file provides for each disputed domain, an integer value representing the risk-level of that domain where the values 1, 2, and 3 represent respectively a low, medium, and high-risk domain.

```
"airbnb.com": {  
  "aribnb.com": 1,  
  "ajrbnb.com": 2,  
  "airbnbb.com": 3,  
  "airbnk.com": 1,  
  "arbnb.com": 1,  
  "airnb.com": 2,  
  "airbbn.com": 2,  
  "airbanb.com": 1  
},
```

The dataset also includes the **zonefile.txt** that contains a snapshot of registered domains as observed in the zone files. Note that you automatically lose money from your budget if you recommend the registration of a domain name that is already registered.

Rubric

Your performance will be evaluated based on the following criteria:

- Performance of the recommendations on the hold-out dataset (30%)
 - The ability of your proposed strategy to reduce the number of disputed domains for the companies in the holdout dataset used for the internal competition will be assessed. This is what is shown on the public leaderboard. The CTO of your firm will also gauge the effectiveness of the strategy relative to the potential profits you can make for *DomainSense*.
- Persuasiveness of sale pitch (20%)
 - You must convince the members of *Strategic Ventures Inc.* of the viability of your proposed strategy using strong and coherent arguments presented in a compelling manner.
- Performance of the recommendations for the prospect's companies (30%)
 - Your proposed strategy will be tested on companies that *Strategic Ventures Inc.* is looking to protect under the new contract. Your performance on this hold-out dataset will be reported to you on the day of your presentation.
- Clarity and technical presentation (20%)
 - Your code and technical report should be presented professionally and with enough details to assist the customer acquisition team with submitting a strong proposal to the customer.

Submitting Recommendations to the Leaderboard

To measure the effectiveness of your approach on the validation dataset, submit your recommendations for defensive registrations by selecting the *defensive registration assignment* at <https://riposte-stu-20.storm.gatech.edu/assignments>. The file you submit must end in a `.txt` extension, and each line within that file must be formatted as `company_domain:recommendation` pair. The file should contain the ordered list of domains you recommend for registration. During the evaluation, any registration that does not fit within the budget will be discarded. For instance, if under the \$100,000 budget, you recommend defensively registering more than the maximum of 500 domains, only the first 500 domains in the list will be considered in the scoring.

An example of the live leaderboard is shown below:

Defensive Submissions Leaderboard												
Top submission for each (user, tag) pair ordered by total disputes avoided (low+medium+high).												
#	User	Tag	Disputes Avoided			Disputes Encountered			Financial Overview			Submitted
			Low Risk	Medium Risk	High Risk	Low Risk	Medium Risk	High Risk	Registration Costs (\$)	Penalties (\$)	Profits (\$)	
1	cautious capitalists	PitchReady	88	20	0	22	6	3	-25,000.00	-12,900.00	+62,100.00	2025-10-31 14:44:14
2	risk masters	PitchReady	33	24	1	77	2	2	-40,000.00	-26,100.00	+33,900.00	2025-10-31 14:49:25
3	budget burners	Prototype	27	2	0	83	24	3	-100,000.00	-42,000.00	0.00	2025-10-31 14:46:21
4	start praying	Experimental	0	0	0	110	26	3	0.00	-51,300.00	+48,700.00	2025-10-31 14:47:50

To help you get started, the notebook `helper.ipynb` implements the following features:

- A **Company** class that holds details about a Fortune 500 company, and a **CompanyLoader** class that combines the information about all the companies in a dataset (e.g., `companies-training.jsonl`) into an accessible interface for retrieval. *In general, you should not modify those classes, and their functions.*
- A **BrandProtector** class that simulates an online brand protection service provider, which you represent in this assignment. The class contains the following functions:
 - A constructor (`__init__`) that sets the cost of various actions (i.e., domain registration, domain disputes) and loads the companies and disputes data into memory. *We do not anticipate any need for you to modify this piece of code.*

- A function (**generate**) that applies the five typo-squatting models of Wang et al. [1] to the companies loaded in memory and generates domains that are available for registration. Although ***you can extend this function to incorporate other transformations***, always make sure that it only returns domain names that are not registered (i.e., not in the attached zone file) as recommending registered domains will cost you money. This check is implemented by default.
- A placeholder function (**predict**) that demonstrates how to generate the recommendations and produce the output in the expected format. **You must modify this function to implement your team's strategy for recommending domain names that should be defensively registered**, while adhering to the output format. Use whatever strategy you deem best to allocate your budget across the different companies.
- The **score** function that reads your recommendations from a text file and computes the relevant metrics such as the number of disputes avoided, the number of disputes encountered, your expenses in terms of domain registrations and penalties, and the profits you made. This is there to aid you with testing. Given that the data does not include the disputes encountered by the companies in the validation set, this function will only work for the training dataset. **Do not modify this function.**

Final Solution Submission Instructions

The day before the in-class presentation (i.e., **December 1st, 2025 at 06:00 pm**), you must upload your Python code to Canvas, with instructions on how to run it to produce the desired output given an input file in the specified format.

References

[1] Wang, Yi-Min, Doug Beck, Jeffrey Wang, Chad Verbowski, and Brad Daniels. "Strider Typo-Patrol: Discovery and Analysis of Systematic Typo-Squatting." *Proceedings of the 2nd Workshop on Steps to Reducing Unwanted Traffic on the Internet*, USENIX Association, July 2006, San Jose, CA.